



What is Data Binding in Angular?

Definition

Data Binding in Angular is a mechanism that allows synchronization of data between the component (TypeScript) and the view (HTML template).

It means:

- Data can flow from component → template
- Data can flow from template → component
- Or both ways

Angular uses binding syntax to connect the UI and business logic.





Types of Data Binding in Angular

Type	Direction
Interpolation	Component → Template
Property Binding	Component → Template
Event Binding	Template → Component
Two-Way Binding	Both directions





Interpolation ({{ }})

Interpolation is used to display component data in the HTML template.

Key Points:

- ✓ One-way binding
- ✓ Data flows from component → view

✓ Example

```
export class AppComponent {  
  username = 'Satendra';  
  age = 28;  
}
```

```
<h2>Hello {{ username }}</h2>
```

```
<p>Age: {{ age }}</p>
```





Property Binding ([property])

Property binding is used to bind a component property to an HTML element property.

Key Points:

- ✓ One-way binding
- ✓ Data flows from component → view

✓ Example

```
export class AppComponent {  
  isDisabled = true;  
  imageUrl = 'assets/angular.png';  
}
```

```
<button [disabled]="isDisabled">Submit  
</button>  
<img [src]="imageUrl">
```





Property Binding vs Interpolation

Interpolation	Property Binding
{{ value }}	[property]="value"
For text	For DOM properties
String only	Any data type





Event Binding ((event))

Event binding is used to handle user actions like click, keyup, submit, etc.

Key Points:

- ✓ One-way binding
- ✓ Data flows from view → component

✓ Example

```
export class AppComponent {  
  count = 0;  
  
  increment() {  
    this.count++;  
  }  
}
```





```
<button(click)="increment()">
```

Increase

```
</button>
```

```
<p>Count: {{ count }}</p>
```

Passing Event Object

```
<input (keyup)="onKeyUp($event)">
```

```
onKeyUp(event: any) {  
  console.log(event.target.value);  
}
```





Two-Way Data Binding ([ngModel])

Two-way binding allows automatic synchronization between component and template.

Key Points:

- ✓ Data flows both ways
- ✓ Combines property + event binding

To use ngModel, you must import FormsModule.

```
import { FormsModule } from '@angular/forms';
```

```
@NgModule({  
  imports: [BrowserModule, FormsModule],  
})
```

```
export class AppModule {}
```





Component (TS)

```
export class AppComponent {  
  username = '';  
}
```

Template (HTML)

```
<input [(ngModel)]="username">  
<p>Entered Name: {{ username }}</p>
```

How Two-Way Binding Works Internally

```
<input  
  [value]="username"  
  (input)="username = $event.target.value">
```





Summary Table

<u>Binding Type</u>	<u>Syntax</u>	<u>Direction</u>
Interpolation	{{ value }}	Component → View
Property Binding	[property]	Component → View
Event Binding	(event)	View → Component
Two-Way Binding	[(ngModel)]	Both





Angular

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Thank you

I hope you find this post helpful.



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